

DevOps Final Project

deployment.yml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-deploy
  namespace: my-bank
  labels:
    name: my-deploy
spec:
  replicas: 1
  selector:
    matchLabels:
      apptype: web-backend
  strategy:
    type: RollingUpdate
  template:
    metadata:
      labels:
        apptype: web-backend
    spec:
      containers:
        - name: my-web
          image: 22csl258/mysimplewebapplication:tagname
          ports:
            - containerPort: 9001
---
apiVersion: v1
kind: Service
metadata:
  name: my-service
  namespace: my-bank
  labels:
    app: my-service
spec:
  type: NodePort
  ports:
    - port: 9001
      targetPort: 8080
```

```
nodePort: 30007
selector:
  apptype: web-backend
```

Minikube

```
info: Adding new user 'grafana' (UID 107) with group 'grafana' ...
info: Not creating home directory '/usr/share/grafana'.
### NOT starting on installation, please execute the following statements to configure grafana to start automatically using systemd
sudo /bin/systemctl daemon-reload
sudo /bin/systemctl enable grafana-server
### You can start grafana-server by executing
sudo /bin/systemctl start grafana-server
Processing triggers for man-db (2.12.0-4build2) ...
Synchronizing state of grafana-server.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable grafana-server
Created symlink /etc/systemd/system/multi-user.target.wants/grafana-server.service → /usr/lib/systemd/system/grafana-server.service.
• grafana-server.service - Grafana instance
   Loaded: loaded (/usr/lib/systemd/system/grafana-server.service; enabled; vendor preset: enabled)
   Active: active (running) since Sat 2025-03-22 08:18:26 UTC; 31ms ago
     Docs: http://docs.grafana.org
   Main PID: 85935 ((grafana))
     Tasks: 1 (limit: 4597)
    Memory: 244.6M
   CGroup: /system.slice/grafana-server.service
           └─85935 "(grafana)"

Mar 22 08:18:26 LAPTOP-QG4ITHLE systemd[1]: Started grafana-server.service >

pranav@LAPTOP-QG4ITHLE:~$ minikube start
minikube v1.35.0 on Ubuntu 24.04 (amd64)
* Using the docker driver based on existing profile
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.46 ...
* Updating the running docker "minikube" container ...
* Preparing Kubernetes v1.32.8 on Docker 27.4.1 ...
* Verifying Kubernetes components...
  * Using image docker.io/kubernetes/dashboard:v2.7.0
  * Using image docker.io/kubernetes/metrics-scraper:v1.0.8
  * Using image gcr.io/k8s-minikube/storage-provisioner:v5
  * Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

* Enabled addons: default-storageclass, storage-provisioner, dashboard
* Done! kubectrl is now configured to use "minikube" cluster and "default" namespace by default
pranav@LAPTOP-QG4ITHLE:~$
```

```
main PID: 65520 (grafana)
Tasks: 1 (limit: 4597)
Memory: 244.6M ()
CGroup: /system.slice/grafana-server.service
└─65535 "grafana"
```

Mar 22 08:10:26 LAPTOP-QG4ITHLE systemd[1]: Started grafana-server.service

```
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pranav@LAPTOP-QG4ITHLE:~$ minikube service my-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	my-service	7890	http://192.168.49.2:7890

```
! Starting tunnel for service my-service.
```

NAMESPACE	NAME	TARGET PORT	URL
default	my-service		http://127.0.0.1:42863

```
! Opening service default/my-service in default browser...
! http://127.0.0.1:42863
! Because you are using a Docker driver on linux, the terminal needs to be open to run it.
```

Pipeline script - jenkins

JAVA Application pipeline

```
{
    agent any

    stages { stag
        ('scm'){
            steps {
                git branch: "
            }
        }
        stage('builb-clean') {
            steps {
                sh "mvn clean"
            }
        }
        stage('build-validate') {
            steps {
                sh "mvn validate"
            }
        }
        stage('build-com') {
            steps {
                sh "mvn compile"
            }
        }
    }
}
```

```
stage('build-test'){
    steps {
        sh "mvn test"
    }
}

stage('build-install'){
    steps {
        sh "mvn package"
    }
}

stage('build to images'){
    steps {
        script{
            sh 'docker build -t .'
        }
    }
}

stage('push to hub'){
    steps {
        script{
            withDockerRegistry(credentialsId: 'Docker_cred', url: 'https://index.docker.io/v1/') {
                sh 'docker push '
            }
        }
    }
}
```

```

}

stage('Deploy App'){
    steps {
        withKubeConfig(caCertificate: '', clusterName: 'minikube', contextName: 'minikube',
credentialsId: 'mukubeconfig_011', namespace: '', restrictKubeConfigAccess: false, serverUrl:
'https://192.168.49.2:8443') {
            sh 'kubectl apply -f deployment.yml --validate=false'
        }
    }
}

stage('Test'){
    steps {
        withKubeConfig(caCertificate: '', clusterName: 'minikube', contextName: 'minikube',
credentialsId: 'mukubeconfig_011', namespace: '', restrictKubeConfigAccess: false, serverUrl:
'https://192.168.49.2:8443') {

            sh 'minikube service my-service --url | xargs curl'
        }
    }
}
}

```


